Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/22/2023
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ara@engr.psu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	814-865-9427
Organization Name:	Pennsylvania State University		
PI Address 1:	Civil and Environmental Engineering		
PI Address 2:	231D Sackett Bldg		
PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/22/2016	**End Date:**	11/21/2023
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Jackson, Enrique	**Contact Phone:**	
Contact Email:	enrique.m.jackson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/21/2023 per NSSC information (Ed., 9/2/23) NOTE: End date changed to 11/21/2022 per NSSC information (Ed., 10/15/21) NOTE: End date changed to 11/21/2021 per NSSC information (Ed., 10/1/20)		
Key Personnel Changes/Previous PI:	Two graduate students finalized their PhD thesis and completed PhD degrees.		
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement's material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments satisfied two significant objectives. First, the microstructure and material properties of the benchmark samples were evaluated; different responses to thermal and mechanical loading were characterized in detail. Any positive attributes found in the unique environment were reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or use can have huge implications on the sustainability of the cement industry.
Task Progress:	All research tasks were completed and a significant amount of additional research was performed via additional NASA funding mechanisms (Physical Sciences Informatics / PSI, Cooperative Agreement Notice / CAN, and NASA Space Technology Graduate Research Opportunities / NSTGRO). Please refer to previous yearly reports.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Articles in Peer-reviewed Journals	Collins PJ, Thomas RJ, Radlinska A. "Influence of gravity on the micromechanical properties of portland cement and lunar regolith simulant composites." Cem Concr Res. 2023 Oct 1;172:107232. https://doi.org/10.1016/j.cemconres.2023.107232 , Oct-2023
Articles in Peer-reviewed Journals	Collins PJ, Radlinska A, Rattenbacher B, Schnellenbach-Held M, Sperl M. "ACI Committee on Lunar Concrete launched following recent advancements from International Space Station experiments." Concr. Int. 2023 Mar 1;45(3):39-43. https://www.concrete.org/publications/internationalconcreteabstractsportal.aspx?m=details&i=51738695 , Mar-2023